Regularity in the Calculus of Variations
MA 374A Assignment 2

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1. Let $\Omega \subset \mathbb{R}^n$ be open, bounded and smooth. Let $0 < \alpha < 1$ be a real number and let $F \in C^{0,\alpha}\left(\overline{\Omega}; \mathbb{R}^n\right)$. Let $u \in W_0^{1,2}(\Omega)$ be a weak solution to the following

$$\begin{cases} -\Delta u = \operatorname{div} F & \text{in } \Omega, \\ \quad u = 0 & \text{on } \partial\Omega. \end{cases}$$

Show that $u \in C^{1,\alpha}\left(\overline{\Omega}\right)$.

2. Let $\Omega \subset \mathbb{R}^n$ be open, bounded and smooth. Show that for any vector field $X \in C^{0,\alpha}\left(\overline{\Omega}; \mathbb{R}^n\right)$ can be written as

$$X = \nabla\phi + Y \qquad \text{in } \Omega,$$

where $\phi \in C^{1,\alpha}\left(\overline{\Omega}\right)$ satisfies $\phi = 0$ on $\partial\Omega$ and $Y \in C^{0,\alpha}\left(\overline{\Omega}; \mathbb{R}^n\right)$ is divergence-free in Ω in the sense of distributions.

Remark 1. *This is the Helmholtz decomposition theorem.*